

Key features of parabolas



- 1
 - a Downwards
 - b Upwards
- 2 Upward
- 3 No, because the entire parabola will be above the x -axis. Also, there is no solution for the equation $x^2 + 6 = 0$.
- 4
 - a No
 - b Yes
 - c No
 - d Yes
- 5
 - a $x = 0, 4$
 - b $y = 0$
 - c 4
- 6
 - a Concave up
 - b $y = 9$
 - c $y = 1$
 - d $x = -2$
 - e $x < -2$
- 7
 - a $(-2, 0)$
 - b Vertex: $(-2, 0)$
 - c
 - i False
 - ii True
 - iii True
 - iv True
- 8
 - a 2
 - b 1
- 9 Concave up
- 10
 - a $(-4, 2)$
 - b 2
- 11 $x = 3$
- 12
 - a No
 - b No
 - c Yes
- 13 Quadratic function A



15 $y = 25$

16

a $x = 14, 6$

b $x = 10$

c Concave down

d $y = 16$

17 The x -coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The y -coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x .

18 $x = -3, 2$

19

a False

b True

20 $x = \frac{-3 \pm \sqrt{93}}{6}$

Graphing quadratics

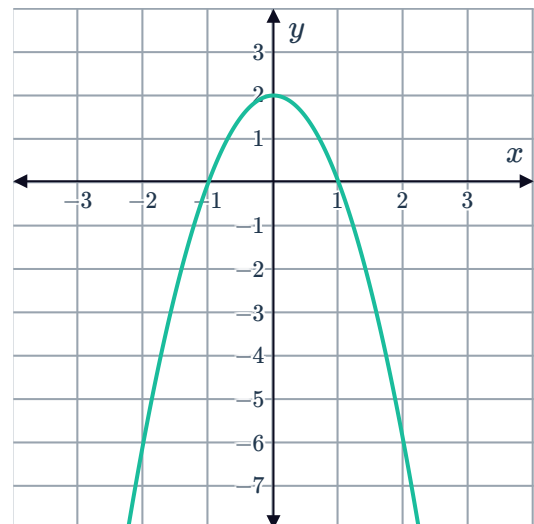
21

a Concave down

c $(0, 2)$

b More steep

d



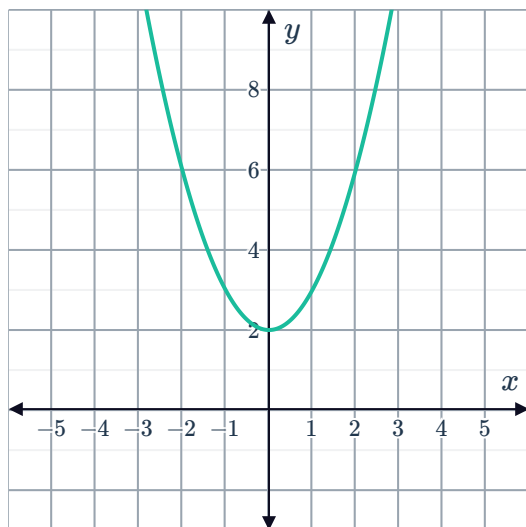
22

a $y = x^2 + 5$

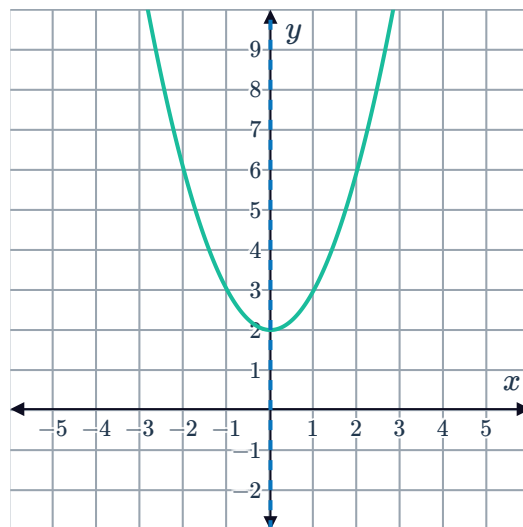
b $x \approx 2.3, -2.3$

23

a



b



c $(0, 2)$

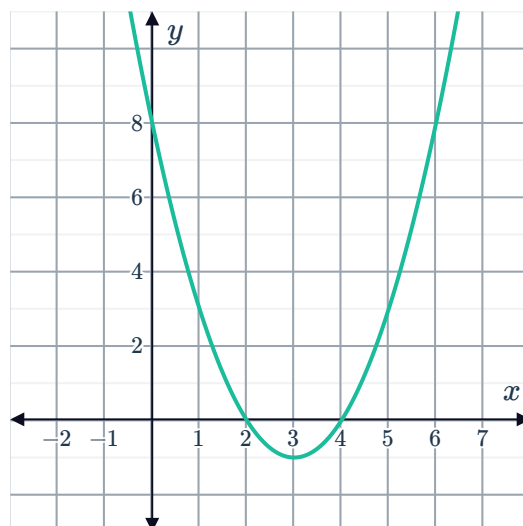
24

a $x = 4, 2$

c $(3, -1)$

b $y = 8$

d



e $x = 4, 2$

25

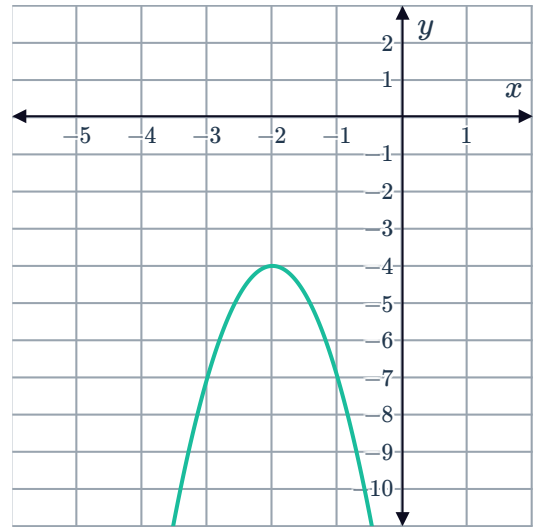
a $(-2, -4)$

b $x = -2$

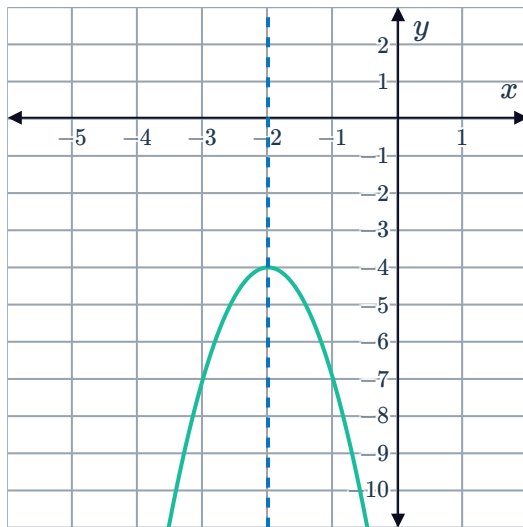
c $y = -7$



d

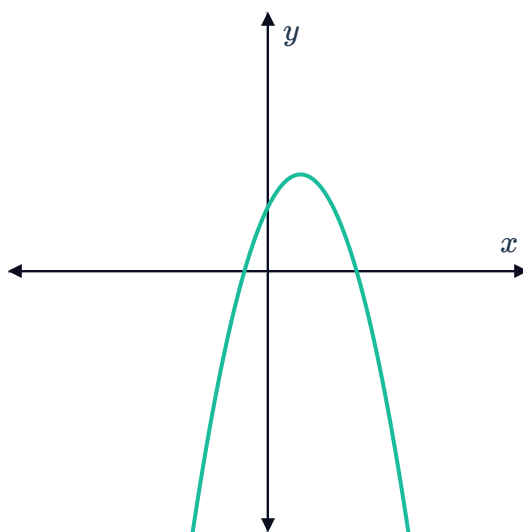


e

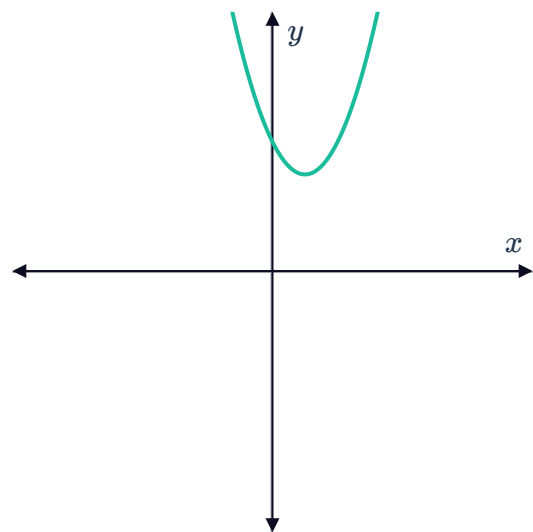


26

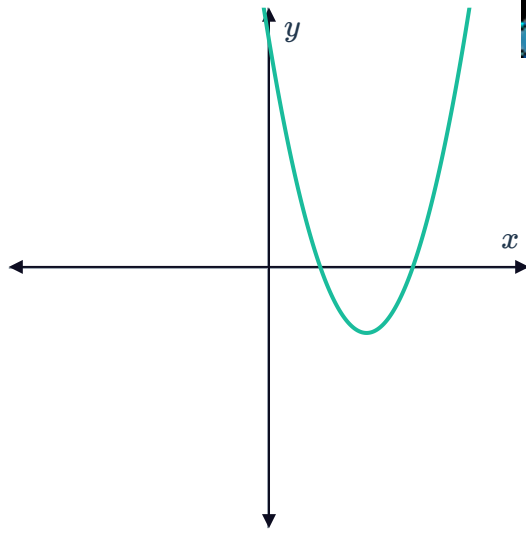
a



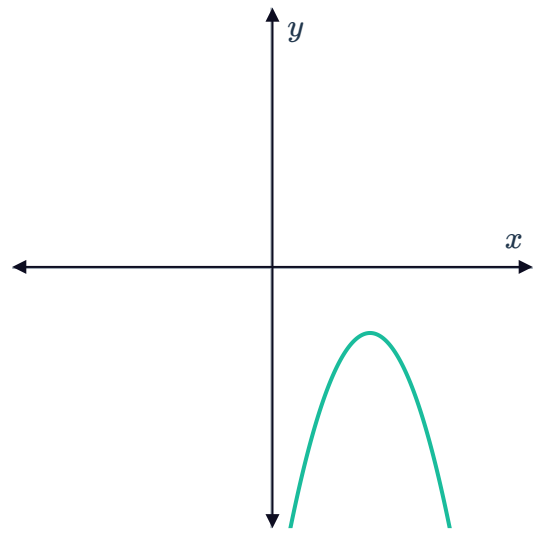
b



c



d



27

a 8

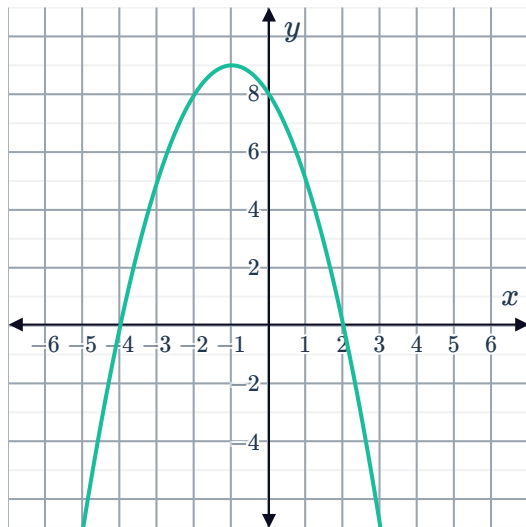
b $x = 2, -4$

c

d $(-1, 9)$

x	-5	-3	-1	1	3
y	-7	5	9	5	-7

e



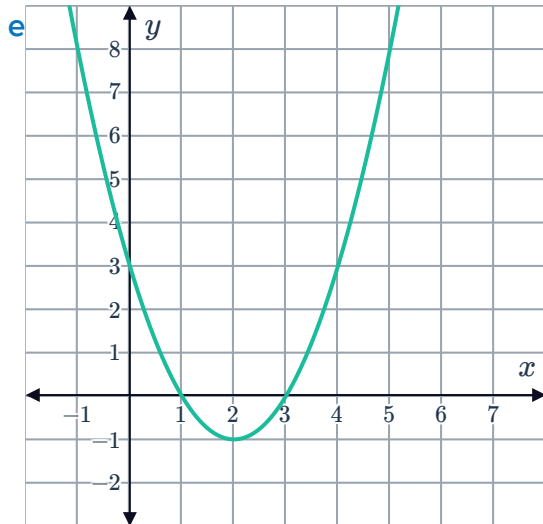
28

a $y = 3$

b $x = 3, 1$

c $x = 2$

d $(2, -1)$



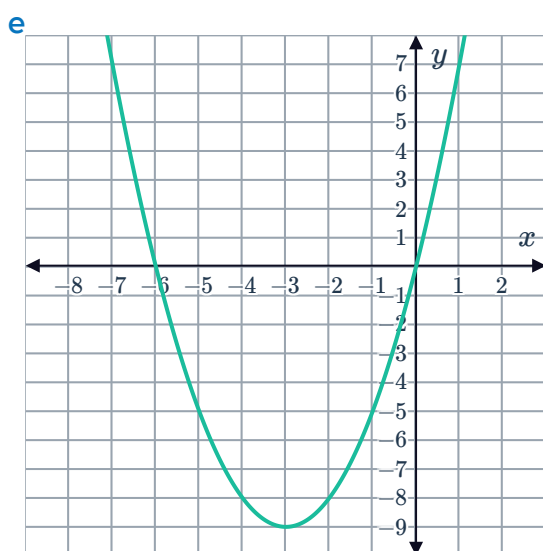
29

a $y = 0$

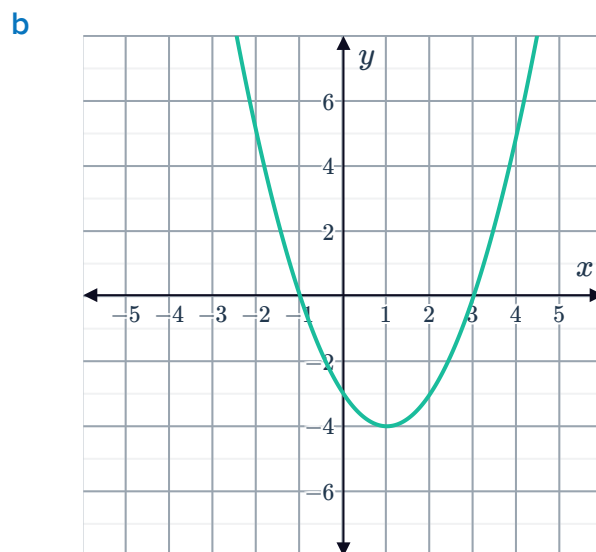
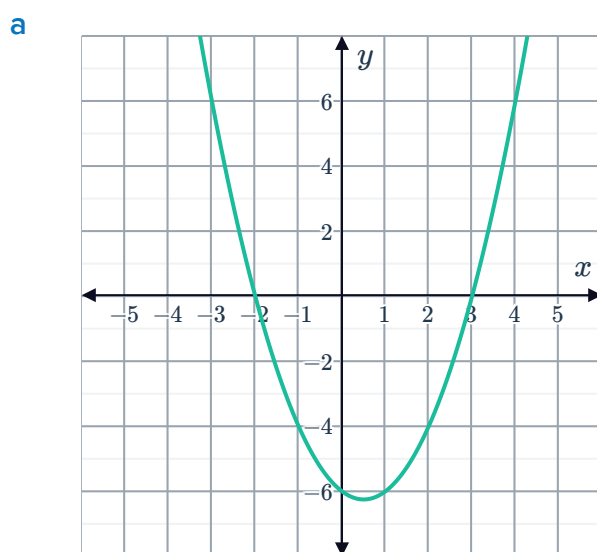
b $x = 0, -6$

c $x = -3$

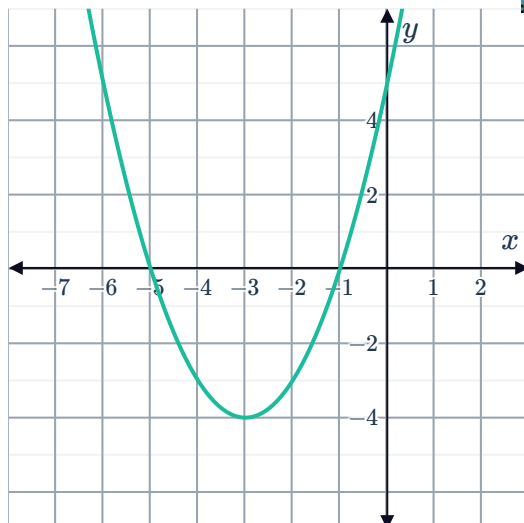
d $(-3, -9)$



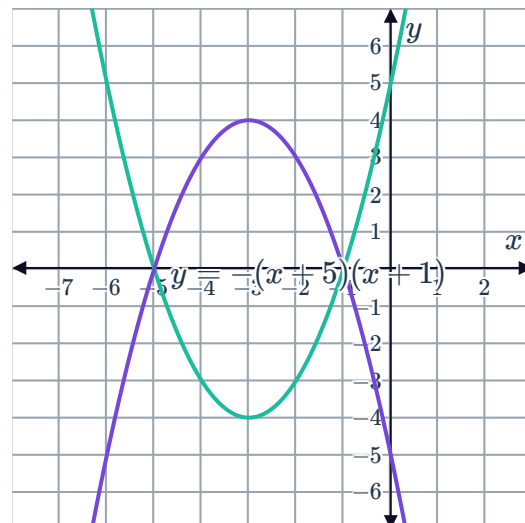
30



31 a



b



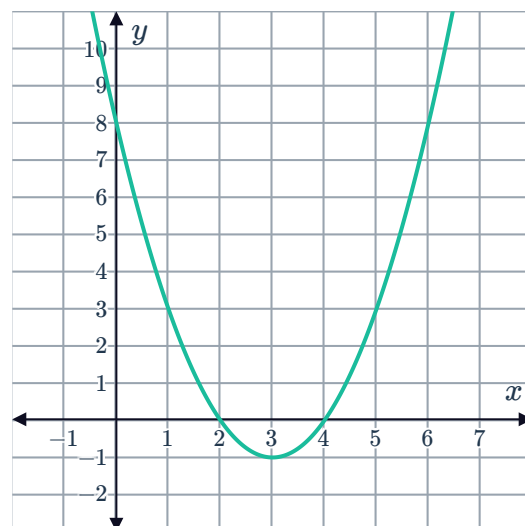
32

a $(x - 2)(x - 4)$

c $(3, -1)$

b $x = 2, 4$

d



33

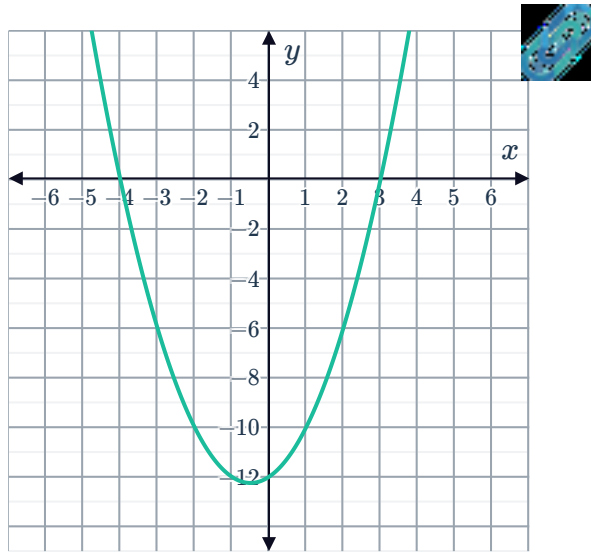
a $x = -4, 3$

c $x = -\frac{1}{2}$

b $y = -12$

d $\left(-\frac{1}{2}, -\frac{49}{4}\right)$

e

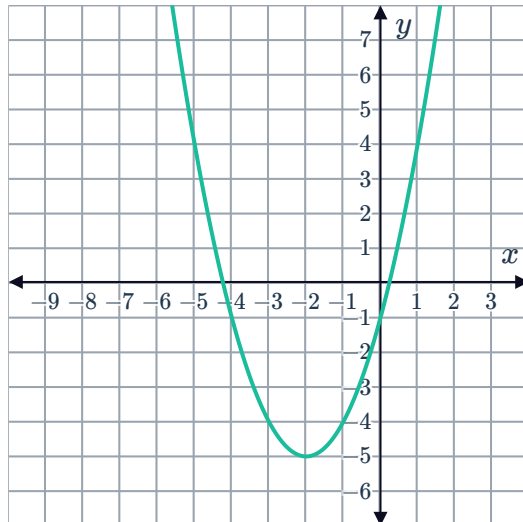


34

a $y = (x + 2)^2 - 5$

c $(-2, -5)$

e



b $y = -1$

d Concave up

f $x \approx 0.2, -4.2$

35

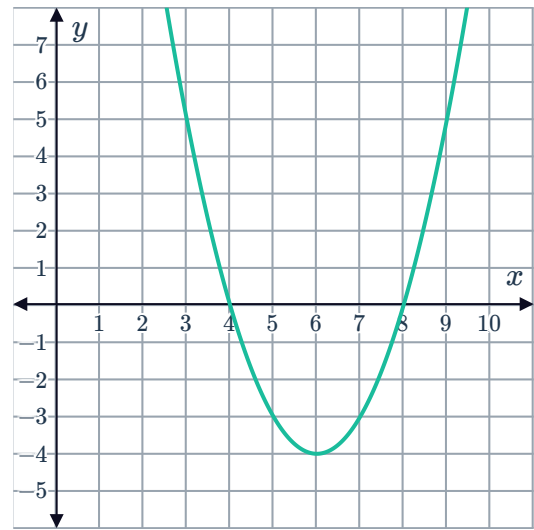
a $x = 8, 4$

b $y = (x - 6)^2 - 4$

c $(6, -4)$



d



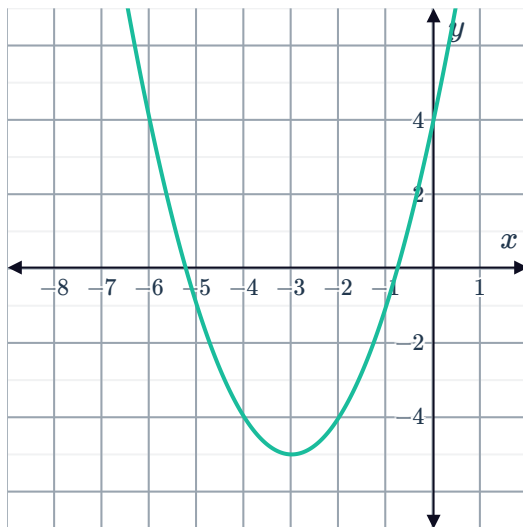
36

a $x = -3$

b -5

c

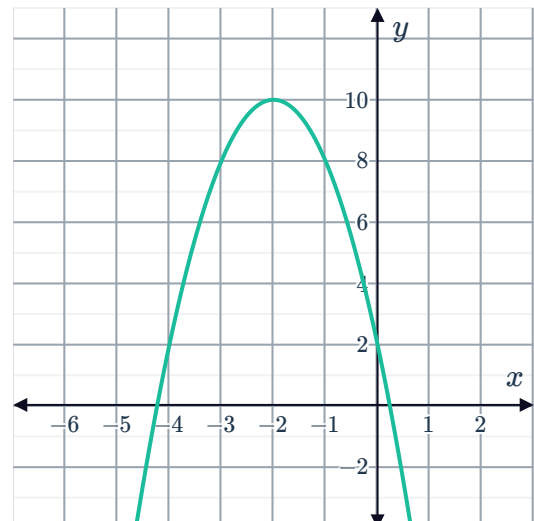
d $x \approx -0.8, -5.2$



37

a $(-2, 10)$

b



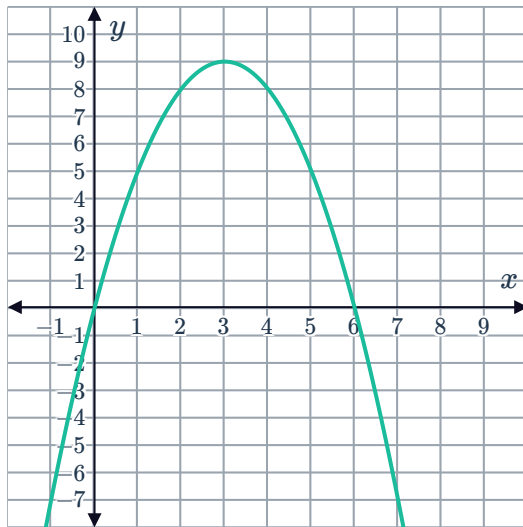


38

a $x = 0, 6$

b $(3, 9)$

c



39

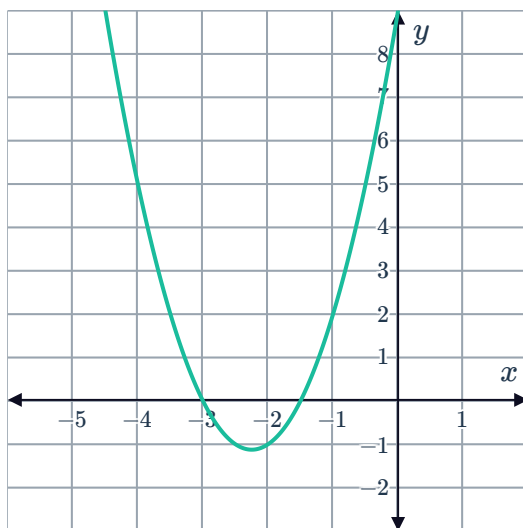
a $x = -\frac{3}{2}, -3$

b $y = 9$

c $x = -\frac{9}{4}$

d $y = -\frac{9}{8}$

e



Graphing quadratics using technology

40

a i $(8, 7)$

ii $(0, 263)$

b i $(-6, 4)$

ii $(0, -140)$

a $(0, -12)$

b No



c $x < 0$

42

a $x = 2, 6$

b $y = -24$

c $x = 4$

d $y = 8$